FI 362: Introduction to Financial Econometrics

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Outline

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About me

- Lecturer in Economics at the University of Oxford.
- Research and teaching in microeconomics, economic theory, and econometrics.
- This week: an empirical, hands-on introduction to **financial time series**.

Housekeeping

- Non-alcoholic drinks and snacks are fine; please, no hot/cold meals during lectures.
- All data and the companion R notebook live in the course folder (see the last slides).

Reading Materials

Main Textbook

- Tsay, R. S. (2010). *Analysis of Financial Time Series* (3rd ed.). Wiley Series in Probability and Statistics. ISBN: 978-0470414354

Other reading materials:

- Daníelsson, Jon (2012). *Financial Risk Forecasting: The Theory and Practice of Forecasting Market Risk, with Implementation in R and Matlab*. Print ISBN:9780-470669433; Online ISBN:978-1119205869
- McNeil, A. J., Frey, R., & Embrechts, P. (2015). *Quantitative Risk Management: Concepts, Techniques and Tools* Princeton University Press
- Gouriéroux, C., & Jasiak, J. (2001). Financial econometrics: Problems, models, and methods. Princeton University Press

Outline

Introduction

- Financial time series analysis is concerned with the theory and practice of asset valuation over time.
- Both financial theory and its empirical time series contain an element of uncertainty. For example,
  - volatility is not directly observable
  - $\therefore$ statistical theory and methods play an important role in financial time series analysis
- **Objective:**
  - some knowledge of financial time series
  - introduce some statistical tools useful for analyzing these series
  - gain experience in financial applications of various econometric methods

Today's Learning Outcomes

By the end of **today** you will be able to:

- Compute **simple** and **log** returns from prices.
- Aggregate returns across **time** (compounding) and across **assets** (portfolios).
- Compute and interpret the **sample moments** of returns (mean, sd, skewness, kurtosis).
- Test **normality** of returns using skewness, kurtosis, and the Jarque–Bera test.
- Recognise the empirical **stylised facts** of returns: fat tails, skewness, and volatility clustering.

Where We Are Going (Course Outcomes)

Over the whole course you will be able to:

- Construct and interpret ARMA and GARCH models to analyze financial returns.
- Forecast market volatility using econometric techniques.
- Model derivative pricing using continuous-time stochastic processes.
- Apply EVT and VaR techniques to assess market risk and tail behavior.
- Analyze high-frequency data using duration-based models.
- Communicate empirical findings clearly through written and oral presentation.

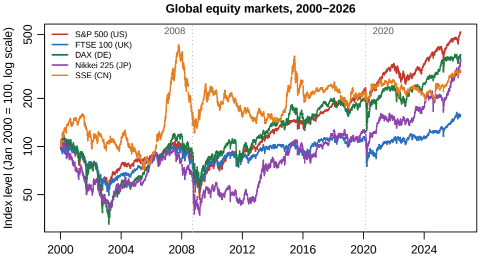
Outline

Motivation: How Risky Is “The Market”?

A question to keep in mind:

- How risky was it to hold equities in 2008? In 2020?
- Do markets around the world move together?
- How would you *measure* “risk” from data like this?

Today we turn these pictures into **numbers**: returns, moments, and tests.



Five global indices, rebased to 100 (log scale). Note the synchronised 2008 and 2020 crashes.

Financial Assets

What is the Financial Asset?

A financial asset is a liquid asset that gets its value from a contractual claim or ownership right, rather than from a physical form with the following characteristics:

- **Intangible Nature**: The value comes from what the asset represents on paper or digitally, not its physical substance.
- **Contractual Claim:** It represents a legal agreement giving you the right to receive cash or ownership in the future.
- **Liquidity:** They are generally much easier and faster to convert into cash compared to physical assets.

Some examples of Financial Assets

Examples:

- **Cash and Cash Equivalents:** Standard currency, bank accounts, or short-term certificates of deposit (CDs).
- **Stocks (Equity):** A fractional share of ownership in a corporation, giving you a claim on a portion of its assets and earnings.
- **Bonds (Debt):** A loan you make to a government or corporation that promises to pay you back with regular interest.
- **Mutual Funds / ETFs**: Bundles of various stocks or bonds grouped together to spread out your risk.

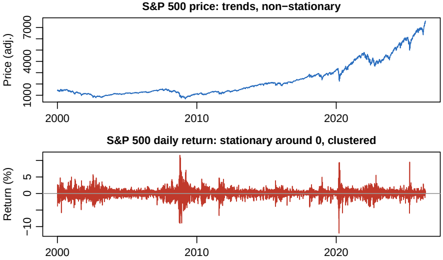
Nota bene: Every financial asset held by an investor shows up as a financial liability or an equity instrument on the balance sheet of the entity that issued it. For example, your bank account is your financial asset, but it is the bank's liability.

Asset Returns: Why Not Prices?

We work with **returns** rather than prices for two reasons:

1. For investors, a return is a complete, scale-free summary of the investment: how much I put in versus how much I get back.
2. Return series have nicer statistical properties than prices (roughly stationary, mean near zero).

Several definitions of “return” exist — we make them precise next.



S&P 500: the price *trends* (non-stationary); the daily return hovers around 0 with bursts of volatility.

Asset Returns - One period

Let P_t be the price of an asset at time index t. Assume for the moment that the asset pays no dividends.

One-period Simple Return
Holding the asset for one period from date $t-1$ to date t would result in a **simple gross return**:

$$P_t = P_{t-1}(1+R_t) \quad \text{or} \quad 1+R_t = \frac{P_t}{P_{t-1}}. \tag{1}$$

Corresponding one-period **simple net return** is

$$R_t = \frac{P_t}{P_{t-1}} - 1 = \frac{P_t - P_{t-1}}{P_{t-1}}. \tag{2}$$

Asset Returns - Multi-period

Let P_t be the price of an asset at time index t. Assume for the moment that the asset pays no dividends.

k-period Simple Return
Holding the asset for k period from date $t-k$ to date t gives a k-period **simple gross return**:

$$\begin{aligned} 1 + R_t[k] = \frac{P_t}{P_{t-k}} &= \frac{P_t}{P_{t-1}} \cdot \frac{P_{t-1}}{P_{t-2}} \cdot \frac{P_{t-2}}{P_{t-3}} \cdot \dots \cdot \frac{P_{t-k+1}}{P_{t-k}} \\ &= (1+R_t)(1+R_{t-1}) \dots (1+R_{t-k+1}) \\ &= \prod_{j=0}^{k-1} (1+R_{t-j}). \end{aligned}$$

k-period simple gross return is just the product of the k one-period simple gross returns involved. This is called a **compound return**. The k-period simple net return is

$$R_t[k] = \frac{P_t - P_{t-k}}{P_{t-k}}.$$

Asset Returns - Multi-period

- In practice, the actual time interval matters for comparing returns (e.g., monthly return or annual return). If no time interval is given, it is implicitly assumed to be one year.
- If the asset was held for k years, then the annualized (average) return is defined as

$$\text{Annualized}\{R_t[k]\} = \left[\prod_{j=0}^{k-1}(1+R_{t-j})\right]^{\frac{1}{k}} - 1.$$

- Geometric mean of the k one-period simple gross returns involved and can be computed by

$$\text{Annualized}\{R_t[k]\} = \exp\left[\frac{1}{k}\sum_{j=0}^{k-1}\ln(1+R_{t-j})\right] - 1.$$

- Approximated the annualized return

$$\boxed{\text{Annualized}\{R_t[k]\} \approx \frac{1}{k}\sum_{j=0}^{k-1} R_{t-j}.} \tag{3}$$

Quick Check: Does 50% Down then 50% Up Break Even?

Your turn: think–pair–share

A stock falls **50%** one period, then rises **50%** the next. Are you back to where you started? Discuss with your neighbour, then we reveal.

- Gross returns *multiply*: $(1-0.5)(1+0.5) = 0.75$, so you are down 25%, **not** even.
- Simple returns are not additive across time — compounding matters.
- In **log returns**: $\ln(0.5) + \ln(1.5) = -0.693 + 0.405 = -0.288 < 0$. Log returns *do* add, and the sum is negative — consistent with a loss.

This is exactly why we introduce log returns next.

Compounding Interest Rate: Example

Table: Illustration of Effects of Compounding: Time Interval Is 1 Year and Interest Rate Is 10% per Annum

Type	Number of Payments	Interest Rate per Period	Net Value
Annual	1	0.1	\$ 1.10000
Semi-annual	2	0.05	\$ 1.10250
Quarterly	4	0.025	\$ 1.10381
Monthly	12	0.1/12	\$ 1.10471
Weekly	52	0.1/52	\$ 1.10506
Daily	365	0.1/365	\$ 1.10516
Continuously	∞	—	\$ 1.10517

Continuous Compounding

- As the number of compounding periods $m \to \infty$, $\left(1 + \frac{r}{m}\right)^m \to e^r$.
- The net value A of **continuous compounding** is

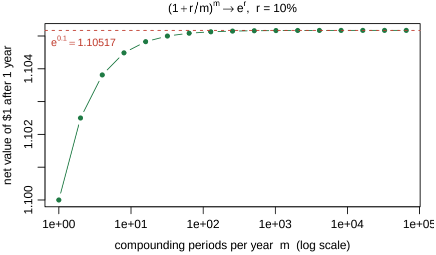
$$A = C \exp(r \times n), \qquad (4)$$

where r is the interest rate per annum, C the initial capital, n the number of years.

- Equivalently,

$$C = A \exp(-r \times n), \qquad (5)$$

the *present value* of A dollars n years from now.



Net value of $1 vs. compounding frequency m. The dashed line is $e^{0.1} = 1.10517$.

Continuously Compounded Return (Log Return)

- The natural logarithm of the simple gross return is the continuously compounded return or *log return*:

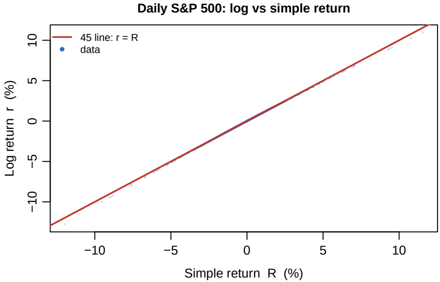
$$r_t = \ln(1 + R_t) = \ln \frac{P_t}{P_{t-1}} = \ln P_t - \ln P_{t-1} = p_t - p_{t-1}. \tag{6}$$

- Log returns are additive across time. For multiperiod returns,

$$r_t[k] = \ln\big(1 + R_t[k]\big) = r_t + r_{t-1} + \cdots + r_{t-k+1} = \sum_{j=0}^{k-1} r_{t-j}.$$

Simple vs. Log Returns: How Different Are They?

- For small moves, $r_t = \ln(1 + R_t) \approx R_t$.
- They diverge only for large returns:
  - big up-move: $r_t < R_t$
  - big down-move: $|r_t| > |R_t|$
- At **daily** frequency the two are nearly identical — so we use whichever is convenient (log, for additivity).



Daily S&P 500. Points hug the 45° line; they peel away only in the tails.

Portfolio Return

- The simple net return of a portfolio of N assets is a weighted average of the simple net returns of the assets, where each weight is the share of portfolio value in that asset.
- Let p be a portfolio that places weight w_i on asset i. The simple return of p at time t is

$$R_{p,t} = \sum_{i=1}^{N} w_i R_{i,t},$$

  where $R_{i,t}$ is the simple return of asset i. (Exact.)
- For the continuously compounded (log) return of the portfolio we use the approximation

$$\boxed{r_{p,t} \approx \sum_{i=1}^{N} w_i r_{i,t}}$$

  which is excellent at daily frequency (verified in the companion notebook).

Dividend Payment & Excess Return

- If an asset pays dividends periodically, we must modify the definitions of asset returns.
- Let D_t be the dividend paid between dates $t-1$ and t, and P_t the price at the end of period t:

$$R_t = \frac{P_t + D_t}{P_{t-1}} - 1 \qquad \Rightarrow \qquad r_t = \ln(P_t + D_t) - \ln(P_{t-1}).$$

- **Excess return** of an asset at time t is the difference between the asset's return and the return on some reference asset, e.g. a short-term U.S. Treasury bill:

$$Z_t = R_t - R_{0,t} \qquad z_t = r_t - r_{0,t}.$$

Long vs Short Position

- A **long** position means owning the asset: you profit if the price rises.
- A **short** position means selling an asset you do not own: you borrow it from a holder, sell it, and are obliged to buy back the same number of shares later to return them. You profit if the price falls.
- Key subtlety: you repay equal shares, not equal dollars.

Mini-example

Short 100 shares at $50 (receive $5,000). If the price falls to $40, buy back for $4,000: profit $1,000. If instead it rises to $65, buy back for $6,500: loss $1,500. Losses on a short are, in principle, unbounded.

Summary of Relationship

- The relationships between simple return R_t and continuously compounded (or log) return r_t are

$$r_t = \ln(1 + R_t), \qquad R_t = e^{r_t} - 1$$

- If the returns are in percentages (%), then

$$r_t = 100 \cdot \ln\left(1 + \frac{R_t}{100}\right), \qquad R_t = 100(e^{\frac{r_t}{100}} - 1)$$

Example 1.1

- If the monthly log return of an asset is 4.46%, corresponding monthly simple return is $100\,[\exp(4.46/100) - 1] = 4.56\%$
- If the monthly log returns of an asset within quarter are $4.46\%, -7.34\%, 10.77\%$, then quarterly log return of the asset is the sum of them, 7.89%.

Outline

Why Distributions? A Bridge

- So far each return is a single number. But returns are random: to reason about risk we need their **distribution**.
- Returns vary in two directions that we must describe jointly:
  - across **assets** (markets move together — recall the 2008/2020 picture);
  - across **time** (calm and turbulent periods — volatility clustering).
- This motivates three tools we build next:
  - the joint distribution (all assets, all dates together),
  - the marginal distribution (one asset on its own),
  - the conditional distribution (returns given today's information).

Distribution Properties of Returns

- Consider a collection of N assets held for T time periods, $t = 1, \ldots, T$.
- For each asset i, let r_{it} be its log return at time t.
- The log returns under study are $\{r_{it}; i = 1, \ldots N; t = 1, \ldots T\}$
- Simple returns: $\{R_{it}; i = 1, \ldots N; t = 1, \ldots T\}$
- Log-excess returns $\{z_{it}; i = 1, \ldots N; t = 1, \ldots T\}$

Statistical Distributions & Their Moments

- Let R^k be the k-dimensional Euclidean space. A point in R^k is denoted by $x \in R^k$.
- Two Random Vectors:
  - $\mathbf{X} = (X_1, \ldots, X_k)$ and $\mathbf{Y} = (Y_1, \ldots, Y_k)$
- Let $P(\mathbf{X} \in A, \mathbf{Y} \in B)$ be the probability that $\mathbf{X}$ is in the subspace $A \subset R^k$ and $\mathbf{Y}$ is in the subspace $B \subset R^q$.
- Both random vectors are assumed to be continuous.

Joint Distribution

- The joint probability density function (PDF) of $\mathbf{X}$ and $\mathbf{Y}$ is a function $f(\mathbf{x}, \mathbf{y}) \geq 0$ such that:

$$P(\mathbf{X} \in A, \mathbf{Y} \in B) = \int_A \int_B f(\mathbf{x}, \mathbf{y})\, d\mathbf{y}\, d\mathbf{x}$$

- The joint PDF satisfies:

$$\int_{R^k} \int_{R^q} f(\mathbf{x}, \mathbf{y})\, d\mathbf{y}\, d\mathbf{x} = 1$$

- The joint cumulative distribution function (CDF) is:

$$F(\mathbf{x}, \mathbf{y}) = P(\mathbf{X} \leq \mathbf{x},\ \mathbf{Y} \leq \mathbf{y}) = \int_{-\infty}^{\mathbf{x}} \int_{-\infty}^{\mathbf{y}} f(\mathbf{u}, \mathbf{v})\, d\mathbf{v}\, d\mathbf{u}$$

- The joint distribution fully characterizes the stochastic relationship between $\mathbf{X}$ and $\mathbf{Y}$.

Marginal Distribution

- The marginal PDF of $\mathbf{X}$ is obtained by integrating out $\mathbf{Y}$:

$$f_{\mathbf{X}}(\mathbf{x}) = \int_{R^q} f(\mathbf{x}, \mathbf{y})\, d\mathbf{y}$$

- Similarly, the marginal PDF of $\mathbf{Y}$ is:

$$f_{\mathbf{Y}}(\mathbf{y}) = \int_{R^k} f(\mathbf{x}, \mathbf{y})\, d\mathbf{x}$$

- The marginal distribution of $\mathbf{X}$ describes the behavior of $\mathbf{X}$ alone, ignoring $\mathbf{Y}$.
- $\mathbf{X}$ and $\mathbf{Y}$ are independent if and only if:

$$f(\mathbf{x}, \mathbf{y}) = f_{\mathbf{X}}(\mathbf{x}) \cdot f_{\mathbf{Y}}(\mathbf{y}) \quad \forall\, \mathbf{x}, \mathbf{y}$$

Conditional Distribution

- The conditional PDF of $\mathbf{Y}$ given $\mathbf{X} = \mathbf{x}$ is defined as:

$$f_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x}) = \frac{f(\mathbf{x}, \mathbf{y})}{f_{\mathbf{X}}(\mathbf{x})}, \quad f_{\mathbf{X}}(\mathbf{x}) > 0$$

- It describes the distribution of $\mathbf{Y}$ after observing $\mathbf{X} = \mathbf{x}$.
- By Bayes' rule, the joint PDF can be decomposed as:

$$f(\mathbf{x}, \mathbf{y}) = f_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x}) \cdot f_{\mathbf{X}}(\mathbf{x}) = f_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) \cdot f_{\mathbf{Y}}(\mathbf{y})$$

- If $\mathbf{X}$ and $\mathbf{Y}$ are independent, then:

$$f_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x}) = f_{\mathbf{Y}}(\mathbf{y})$$

  i.e., knowing $\mathbf{X}$ provides no information about $\mathbf{Y}$.

Moments of a Random Variable

- Let X be a scalar continuous random variable with PDF $f(x)$.
- The k-th moment of X is defined as:

$$\mu'_k = E[X^k] = \int_{-\infty}^{\infty} x^k f(x)\, dx$$

- The mean (first moment) measures the central tendency:

$$\mu = E[X] = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

- The k-th central moment is defined around the mean:

$$\mu_k = E[(X-\mu)^k] = \int_{-\infty}^{\infty} (x-\mu)^k f(x)\, dx$$

Variance, Skewness, and Kurtosis

- Variance (2nd central moment) measures dispersion:

$$\sigma^2 = \mathrm{Var}(X) = E[(X-\mu)^2]$$

- Skewness (3rd standardized moment) measures asymmetry:

$$\mathcal{S} = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] = \frac{\mu_3}{\sigma^3}$$

  A symmetric distribution has $\mathcal{S} = 0$.

- Kurtosis (4th standardized moment) measures tail heaviness:

$$\mathcal{K} = E\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] = \frac{\mu_4}{\sigma^4}$$

- Excess kurtosis $= \mathcal{K} - 3$; the normal distribution has $\mathcal{K} = 3$.
- Financial returns often exhibit positive excess kurtosis (fat tails).

Moments of a Random Sample

- Let $\{x_1, \ldots, x_T\}$ be a random sample of X with T observations.
- The sample mean is:

$$\hat{\mu}(x) = \bar{x} = \frac{1}{T}\sum_{t=1}^{T} x_t$$

- The sample variance is:

$$\hat{\sigma}^2(x) = \frac{1}{T-1}\sum_{t=1}^{T}(x_t - \hat{\mu}(x))^2$$

- The sample skewness and sample kurtosis are:

$$\hat{\mathcal{S}}(x) = \frac{1}{(T-1)\hat{\sigma}^3}\sum_{t=1}^{T}(x_t - \hat{\mu}(x))^3, \quad \hat{\mathcal{K}}(x) = \frac{1}{(T-1)\hat{\sigma}^4}\sum_{t=1}^{T}(x_t - \hat{\mu}(x))^4$$

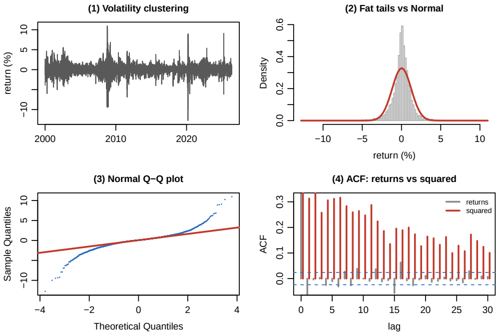
Empirical Moments: Global Markets

Daily % log returns, Jan 2000 – May 2026 (companion data):

Series	Mean	SD	Skew.	Ex. Kurt.
S&P 500 (US)	0.028	1.22	-0.34	10.9
FTSE 100 (UK)	0.004	1.13	-0.37	8.5
DAX (DE)	0.015	1.41	-0.17	6.1
Nikkei 225 (JP)	0.006	1.44	-0.57	6.8
SSE (CN)	0.012	1.42	-0.34	5.7

- Mean daily returns are tiny; standard deviations dominate ($\approx$ 1–1.5%/day).
- Skewness is negative across markets (crashes are sharper than rallies).
- Excess kurtosis is large and positive everywhere — the hallmark of **fat tails**.

The Stylised Facts of Returns



Daily S&P 500 log returns:

1. Volatility clustering: turbulent and calm periods bunch together.
2. Fat tails: the empirical density is taller and has heavier tails than the fitted normal.
3. Q–Q plot bends away from the line in the tails.
4. Serial dependence in variance: returns are nearly uncorrelated, but *squared* returns are strongly autocorrelated.

These motivate the ARCH/GARCH models later in the course.

Asymptotic Distribution Under Normality

- Under the normality assumption, $\hat{\mathcal{S}}(x)$ and $\hat{\mathcal{K}}(x) - 3$ are asymptotically normal:

$$\hat{\mathcal{S}}(x) \xrightarrow{d} \mathcal{N}\left(0,\, \frac{6}{T}\right), \qquad \hat{\mathcal{K}}(x) - 3 \xrightarrow{d} \mathcal{N}\left(0,\, \frac{24}{T}\right)$$

(Snedecor & Cochran, 1980, p. 78)

- These results are the basis for normality tests of asset returns.

Testing Skewness and Kurtosis

- Given a return series $\{r_1, \ldots, r_T\}$, test skewness:

$$H_0 : \mathcal{S}(r) = 0 \quad \text{vs.} \quad H_a : \mathcal{S}(r) \neq 0, \qquad t = \frac{\hat{\mathcal{S}}(r)}{\sqrt{6/T}}$$

  Reject H_0 if the p-value $< \alpha$.

- Test excess kurtosis:

$$H_0 : \mathcal{K}(r) - 3 = 0 \quad \text{vs.} \quad H_a : \mathcal{K}(r) - 3 \neq 0, \qquad t = \frac{\hat{\mathcal{K}}(r) - 3}{\sqrt{24/T}}$$

  Reject H_0 if and only if the p-value $< \alpha$.

Jarque–Bera Test for Normality

- Jarque & Bera (1987) combine skewness and kurtosis tests:

$$\text{JB} = \frac{\hat{\mathcal{S}}^2(r)}{6/T} + \frac{[\hat{\mathcal{K}}(r) - 3]^2}{24/T}$$

- Under H_0 of normality, JB $\xrightarrow{d} \chi^2(2)$.
- **Decision rule:** Reject H_0 if the p-value of the JB statistic is less than the significance level α.
- Financial return series frequently reject normality due to fat tails and skewness.

Jarque–Bera in Action: Every Market Rejects Normality

Series	Skew.	Ex. Kurt.	JB stat	p-value
S&P 500 (US)	-0.34	10.9	31,719	< 0.0001
FTSE 100 (UK)	-0.37	8.5	19,894	< 0.0001
DAX (DE)	-0.17	6.1	10,313	< 0.0001
Nikkei 225 (JP)	-0.57	6.8	12,226	< 0.0001
SSE (CN)	-0.34	5.7	8,480	< 0.0001

- For every global index, the JB statistic is in the tens of thousands — far beyond the $\chi^2(2)$ 5% critical value of 5.99.
- The p-values are effectively zero, driven mostly by excess kurtosis.
- **Conclusion:** the normal i.i.d. model is rejected for real returns — everywhere.

Distributions of Returns

- The most general model for the log returns $\{r_{it}; i = 1, \ldots N; t = 1, \ldots, T\}$ is its joint distribution function:

$$F_r\left(r_{11}, \ldots, r_{N1}; r_{12}, \ldots, r_{N2}; r_{1T}, \ldots, r_{NT}; \mathbf{Y}; \theta\right) \tag{7}$$

  where:

  - $\mathbf{Y}$ is a state vector consisting of variables that summarizes which asset returns are determined
  - θ is a vector parameter determines the distribution function $F_r(\cdot)$

Normal Distribution of Returns

- Under the i.i.d. normal assumption, log returns satisfy:

$$r_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu,\, \sigma^2), \qquad f(r_t) = \frac{1}{\sqrt{2\pi}\,\sigma}\exp\!\left(-\frac{(r_t-\mu)^2}{2\sigma^2}\right)$$

- **Why i.i.d. is unrealistic** (we saw all four on the stylised-facts slide):
  - Volatility clustering: large shocks follow large shocks (ARCH/GARCH effects).
  - Fat tails (leptokurtosis): excess kurtosis > 0; the normal underestimates tail risk.
  - Skewness: return distributions are often negatively skewed.
  - Serial dependence: squared returns are autocorrelated — time-varying variance.

Lognormal Distribution of Returns

- If log returns are normal, $r_t = \ln(P_t/P_{t-1}) \sim \mathcal{N}(\mu, \sigma^2)$, then the simple return $R_t = e^{r_t} - 1$ and the price ratio $P_t/P_{t-1} = e^{r_t}$ follow a lognormal distribution:

$$f(x) = \frac{1}{x\,\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0$$

- **Moments of the price ratio** $X = e^{r_t}$**:**

$$\mathbb{E}[X] = e^{\mu + \sigma^2/2} \qquad \mathrm{Var}(X) = e^{2\mu+\sigma^2}\left(e^{\sigma^2} - 1\right)$$

- **Moments of the log return** r_t: $\mathbb{E}[r_t] = \mu$, $\mathrm{Var}(r_t) = \sigma^2$.
- Lognormality ensures limited liability ($P_t > 0$) and is widely used in option pricing (Black–Scholes).

Nonnormal Stable Distributions

- Stable distributions (Lévy–Stable) generalise the normal: a sum of i.i.d. stable variables is again stable with the same index $\alpha_s \in (0, 2]$.
- Characterised by four parameters: stability index α_s, skewness β_s, scale c, location δ.
- **Cauchy distribution** ($\alpha_s = 1$, $\beta_s = 0$):

$$f(x) = \frac{1}{\pi\, c\left[1 + \left(\frac{x-\delta}{c}\right)^2\right]}, \qquad \mathbb{E}[|X|] = \infty, \quad \mathrm{Var}(X) = \infty$$

- **Relevance:** stable laws capture fat tails and skewness. For $\alpha_s < 2$ the variance is infinite, consistent with extreme market moves.
- **Limitation:** infinite variance breaks the mean–variance framework of portfolio optimisation.

Scale Mixture of Normal Distributions

- A scale mixture of normals (SMN) writes a return as a normal with a *random* variance:

$$r_t \mid v_t \sim \mathcal{N}(\mu,\, v_t\, \sigma^2), \qquad v_t \sim G(\cdot),$$

  where $v_t > 0$ is a latent mixing variable from a mixing distribution G.
- Marginalising over v_t:

$$f(r_t) = \int_0^\infty \frac{1}{\sqrt{2\pi\, v\, \sigma^2}} \exp\left(-\frac{(r_t - \mu)^2}{2v\sigma^2}\right) dG(v)$$

- **Special cases:**
  - $v_t \sim \text{Inv-Gamma}(\nu/2, \nu/2) \Rightarrow$ Student-t with ν d.f. (fat tails, finite variance for $\nu > 2$).
  - G degenerate at 1 $\Rightarrow$ the normal (no mixing).
- SMN models deliver excess kurtosis while staying analytically tractable — a natural bridge to GARCH.

Candidate Distributions at a Glance

Distribution	Finite variance?	Fat tails?	Typical use
Normal	Yes	No	Baseline / Black–Scholes
Lognormal	Yes	—	Prices > 0, option pricing
Student-t (SMN)	Yes ($\nu > 2$)	Yes	Returns with excess kurtosis
Stable ($\alpha_s < 2$)	No	Yes	Extreme moves, crashes

Trade-off: heavier tails fit the data better but can sacrifice finite moments and tractability.

Processes Considered

- Beyond the marginal distribution, we also study the volatility process and the behaviour of extreme returns.
- The volatility process concerns the evolution of the conditional variance of returns over time.
    - Volatility is central to option pricing and risk management.
- **Negative** extreme returns matter for risk management (long positions).
- **Positive** extreme returns matter for short positions.

Outline

Key Takeaways

1. **Returns, not prices.** Returns are scale-free and roughly stationary; prices trend.
2. **Two definitions, one idea.** Simple returns *multiply* across time and *add* across assets; log returns *add* across time and *approximately* add across assets. At daily frequency they nearly coincide.
3. **Compounding $\to e$.** Letting the compounding frequency $\to \infty$ gives continuous compounding and the log return.
4. **Returns are not normal.** Real equity returns show fat tails, mild negative skew, and volatility clustering; Jarque–Bera rejects normality for *every* global market we examined.

Next: we build models that respect these facts — ARMA for the mean, ARCH/GARCH for the variance.

Outline

Getting Started: Data & Companion Notebook

Everything you need is in the course folder

- `data/` — daily prices for 11 global indices and 5 US stocks, Jan 2000 – May 2026 (CSV; built by `download_data.R`).
- `Chapter 1/Day01_Supplement` — a step-by-step R notebook that reproduces every concept and both exercises below.

Tip: the notebook code is column-agnostic — change a ticker name (`"SP500"`, `"AAPL"`, ...) to reuse it on any series.

Exercise 1

- Consider the daily stock returns of American Express (AXP), Caterpillar (CAT), and Starbucks (SBUX) in the last 5 years.
  1. Express the simple returns in percentages. Compute the sample mean, standard deviation, skewness, excess kurtosis, minimum, and maximum of the percentage simple returns.
  2. Transform the simple returns to log returns.
  3. Express the log returns in percentages. Compute the sample mean, standard deviation, skewness, excess kurtosis, minimum, and maximum of the percentage log returns.
  4. Test the null hypothesis that the mean of the log returns of each stock is zero. That is, perform three separate tests. Use 5% significance level to draw your conclusion.

Exercise 2

- Consider the monthly stock returns of the S&P composite index from January 2000 to December 2025. Answer the following questions:
  1. What is the average annual log return over the data span?
  2. Assume that there were no transaction costs. If one invested $1.00 on the S&P composite index at the beginning of 2010, what was the value of the investment at the end of 2025?

Data: `data/sp500_monthly.csv`. Worked solutions are in the companion notebook.